Module 8 slides: Nonlinear PK (worked examples)

Module 8 - Nonlinear Pharmacokinetics · Lecture slides, worked examples · 14 Oct 2026

Name _____ Date _____

After receiving a drug that displays nonlinear kinetics at a dose of 6 mg/kg/day for some time, the steady-state plasma concentration in a 72 kg patient was determined to be 10 mg/L. Since this concentration was lower than the desired steady-state concentration, the dose was increased to 8 mg/kg/day. This dose increase resulted in a steady-state plasma concentration of 20 mg/L, which was greater than the desired steady-state concentration. Calculate Vmax, KM, and a dose rate that should achieve a steady-state plasma concentration of 15 mg/L.

Slide 14 (Example)

Part a: calculate KM.

Slide 14 (Example)

Part b: calculate Vmax.

Slide 14 (Example)

Part c: calculate the dose rate, in mg/day, that should achieve 15 mg/L.

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

Slide 14 (Example) Her key

Answer: 10 mg/L

WHAT THE STEM GIVES

R1	6 mg/kg/day	the first dose rate
C_1	10 mg/L	steady state on the first rate
R2	8 mg/kg/day	the increased dose rate
C_2	20 mg/L	steady state on the second rate
weight	72 kg	*not needed for KM*

THE WORKING, ONE LINE AT A TIME

SETUP

$$KM = \frac{R2 - R1}{(R1/C_1) - (R2/C_2)}, \text{ with } R1 = 6 \text{ mg/kg/day}, C_1 = 10 \text{ mg/L}, R2 = 8 \text{ mg/kg/day}, C_2 = 20 \text{ mg/L}$$

Two dose rates in the same patient, each at steady state, share one Vmax and one KM, so the two dose-rate lines combine into this direct-method line for KM.

ALGEBRA

$$KM = \frac{8 - 6}{(6/10) - (8/20)} = \frac{2 \text{ mg/kg/day}}{0.6 - 0.4 \text{ (mg/kg/day)/(mg/L)}} = \frac{2}{0.2}$$

Each R divided by its own C. The bottom is the earlier rate over its concentration minus the later one, which keeps the result positive. The mg/kg/day cancel and mg/L moves to the top.

ROUND

$$KM = 10 \text{ mg/L}$$

KM comes out in concentration units, mg/L, as it must: it is compared with plasma concentrations, so a KM in mg/kg/day signals the units were not cancelled.

> **DOES THE NUMBER MAKE SENSE?**
>
> A one-third rise in dose doubled the concentration, so the drug is near saturation; KM is a concentration, here equal to the lower measured level: 10 mg/L.

Slide 14 (Example) Her key

Answer: 12 mg/kg/day

WHAT THE STEM GIVES

KM	10 mg/L	from part a, added to C_{ss}
R2	8 mg/kg/day	the dose rate used
C_2	20 mg/L	its own steady state
R1, C_1	6 mg/kg/day, 10 mg/L	the check pair

THE WORKING, ONE LINE AT A TIME

SETUP

$$\text{Vmax} = \frac{R(KM + C_{ss})}{C_{ss}}$$

The dose-rate line R = Vmax × C_{ss} over (KM + C_{ss}), multiplied through by (KM + C_{ss}) and divided by C_{ss}. Any one dose rate with its own steady-state concentration will do.

ALGEBRA

$$\text{Vmax} = \frac{(8\ \text{mg/kg/day})(10 + 20)\ \text{mg/L}}{20\ \text{mg/L}} = \frac{240}{20} = 12\ \text{mg/kg/day}$$

The second pair, 8 mg/kg/day with 20 mg/L. mg/L cancels and Vmax keeps the units of R, mg/kg/day, because the doses were given per kilogram.

ALGEBRA

$$\text{Check: } \frac{(6\ \text{mg/kg/day})(10 + 10)\ \text{mg/L}}{10\ \text{mg/L}} = 12\ \text{mg/kg/day}$$

The first pair, 6 mg/kg/day with 10 mg/L, gives the same 12 mg/kg/day, as it must: both rates come from the same patient with one Vmax. Mixing one pair's R with the other pair's C_{ss} gives a wrong value.

> **DOES THE NUMBER MAKE SENSE?**
> Vmax must exceed both dose rates, because each produced a finite steady state: 12 is above 8 mg/kg/day.

Slide 14 (Example) Her key

Answer: 518.4 mg/day

WHAT THE STEM GIVES

KM	10 mg/L	from part a
Vmax	12 mg/kg/day	from part b
C_{ss}	15 mg/L	the target
weight	72 kg	turns mg/kg/day into mg/day

THE WORKING, ONE LINE AT A TIME

SETUP

$$R = \frac{\text{Vmax} \times C_{ss}}{KM + C_{ss}}$$

With Vmax and KM now known for this patient, the dose-rate line gives the daily rate that holds the steady state at the target of 15 mg/L.

ALGEBRA

$$R = \frac{(12\ \text{mg/kg/day})(15\ \text{mg/L})}{(10 + 15)\ \text{mg/L}} = \frac{180}{25} = 7.2\ \text{mg/kg/day}$$

mg/L cancels and R is per kilogram, because Vmax is. 7.2 mg/kg/day sits between the 6 and 8 mg/kg/day rates, as 15 mg/L sits between 10 and 20 mg/L.

UNIT

R = 7.2 mg/kg/day × 72 kg = 518.4 mg/day

The 72 kg weight turns the per-kilogram rate into a daily dose for this patient. The weight is used only here; KM and Vmax were found without it.

ROUND

R = 518.4 mg/day, about 520 mg/day to the nearest 10

518.4 mg/day is accepted as it stands. When a stem asks for the nearest 10, and also for the nearest 5, it becomes 520 mg/day.

> **DOES THE NUMBER MAKE SENSE?**
> 15 mg/L lies between 10 and 20 mg/L, so the rate must lie between 6 and 8 mg/kg/day: 7.2 mg/kg/day, or 518.4 mg/day for 72 kg.

Her worked answer writes 518.4 mg/day ≈ 520 mg/day; both are accepted.
